

Project Overview

This report presents a full exploratory data analysis of the Zomato Bangalore restaurant dataset containing 8,900+ restaurants across 47+ areas. The analysis covers data cleaning, SQL querying, visualization, and statistical insights to understand restaurant trends, pricing patterns, cuisine popularity, and customer ratings across Bangalore.

Key Statistics

Total Restaurants Analyzed	8,921
Unique Areas	168
Unique Cuisines	101
Average Dinner Rating	3.65 / 5.0
Average Cost for Two	Rs 340
Restaurants with Home Delivery	8,902 (99.8%)
Veg Only Restaurants	645 (7.2%)

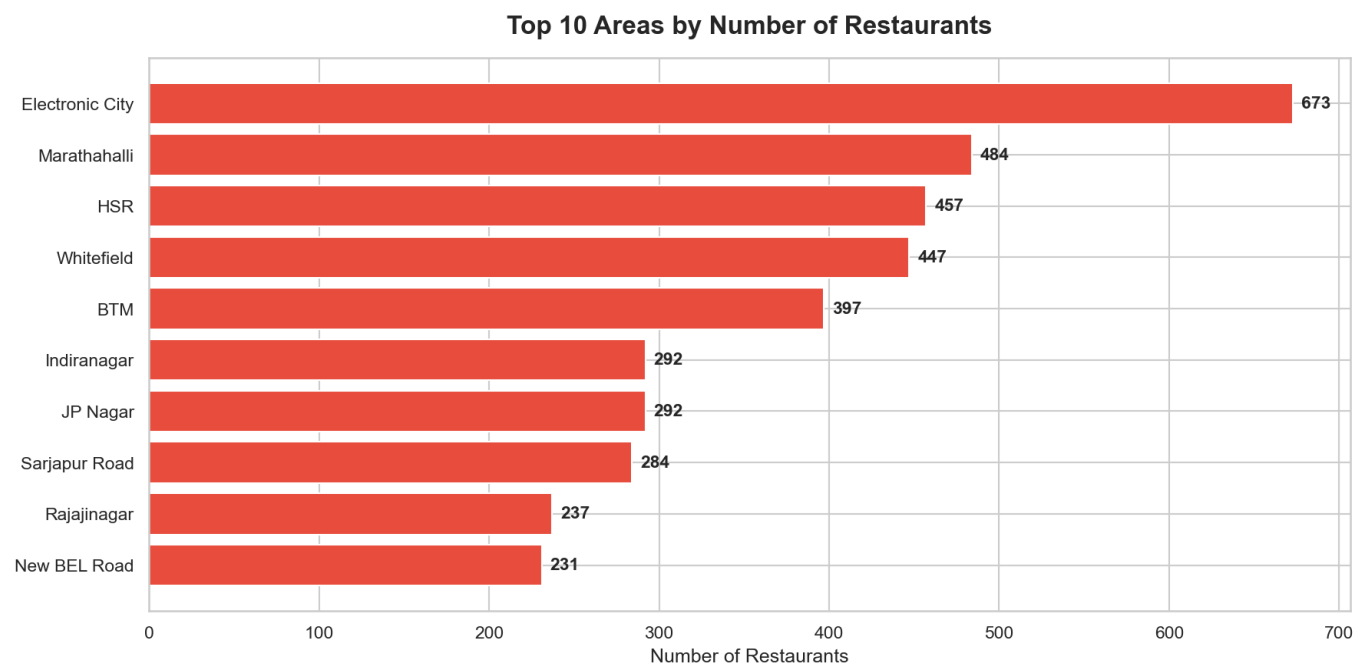
Data Cleaning Summary

1. Removed URL and PhoneNumber columns (not useful for analysis).
2. Removed duplicate restaurants based on Name and Full Address.
3. Cleaned Area column by removing city suffix (Indiranagar, Bangalore -> Indiranagar).
4. Fixed encoding issues in Timing column.
5. Converted rating and cost columns to numeric types.
6. Filled missing text fields with "Not Available".
7. Added 3 new feature columns: Cuisine_Count, Rating_Category, Price_Category.

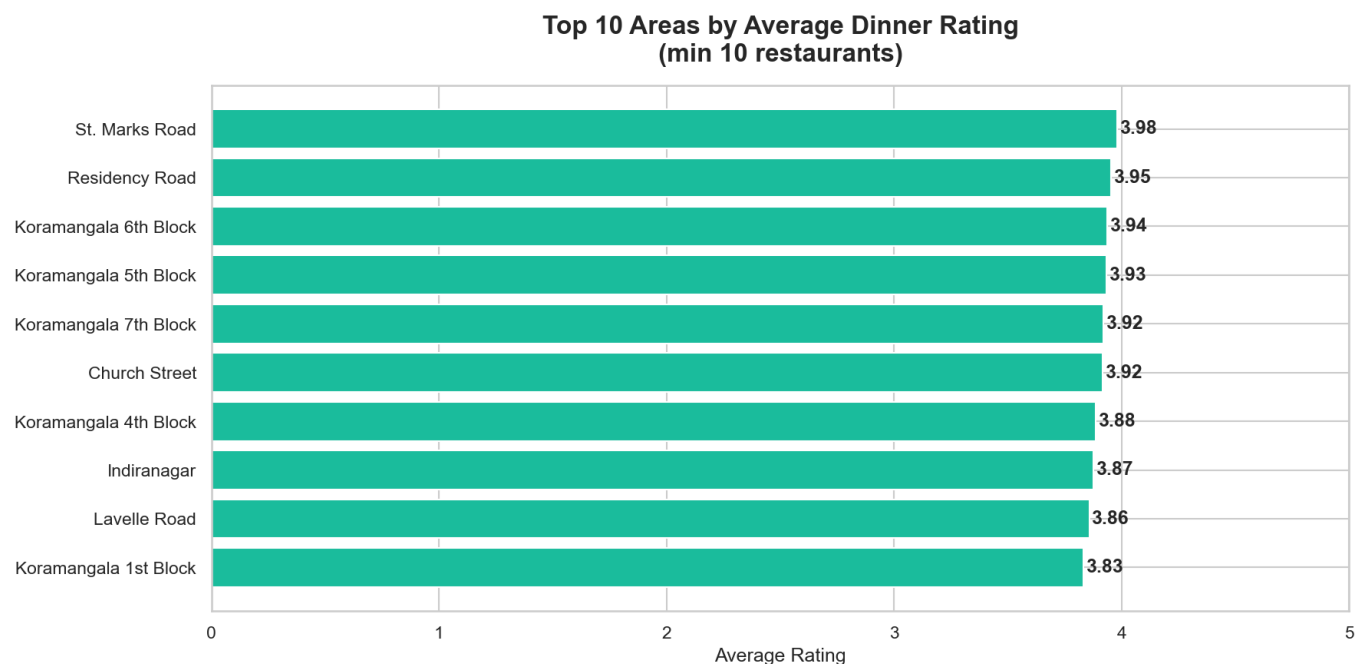
Location Analysis

Whitefield, BTM Layout, and Koramangala are the top 3 areas by number of restaurants. These areas are major IT hubs and commercial zones, which explains the high restaurant density.

Top 10 Areas by Restaurant Count



Top 10 Areas by Average Dinner Rating



Rating Analysis

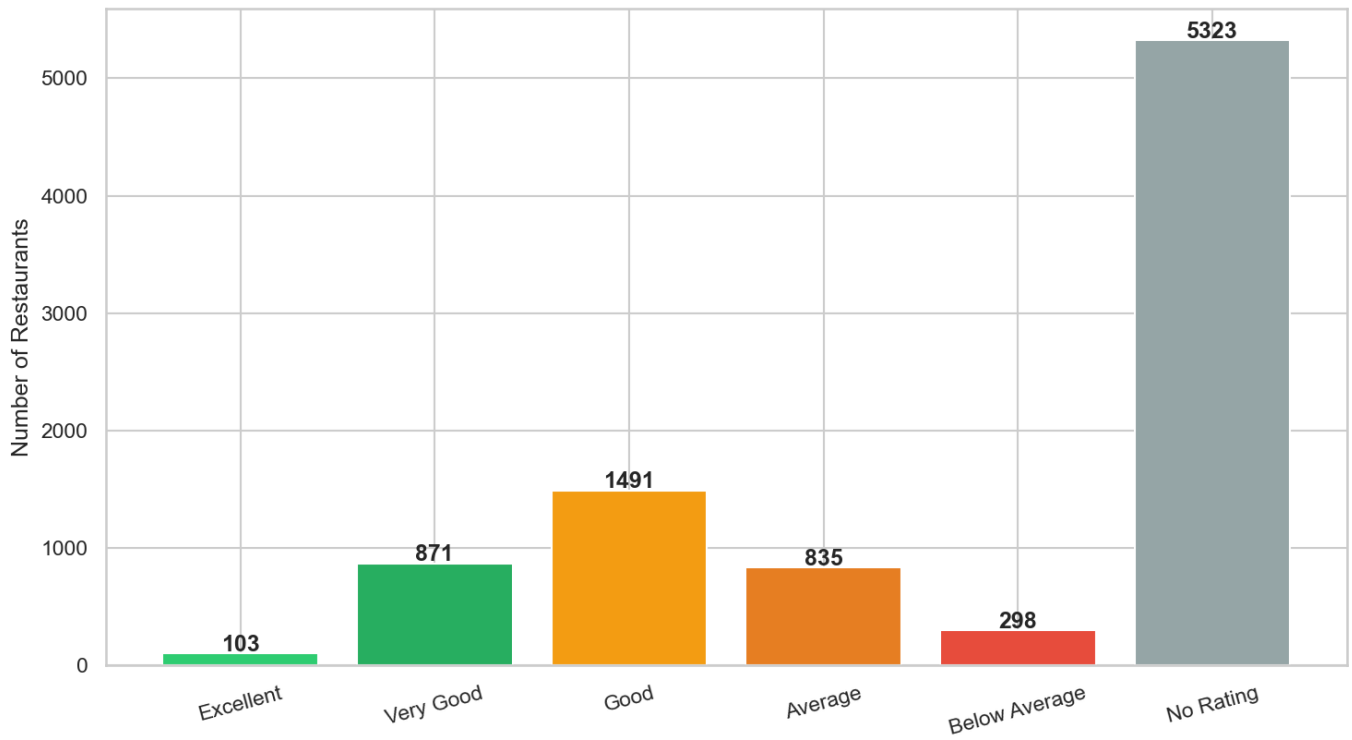
The average dinner rating across all restaurants is 3.65. Most restaurants fall in the 3.5 to 4.5 range, indicating generally positive customer experiences. Only a small percentage are rated below 3.0.

Dinner Rating Distribution



Restaurants by Rating Category

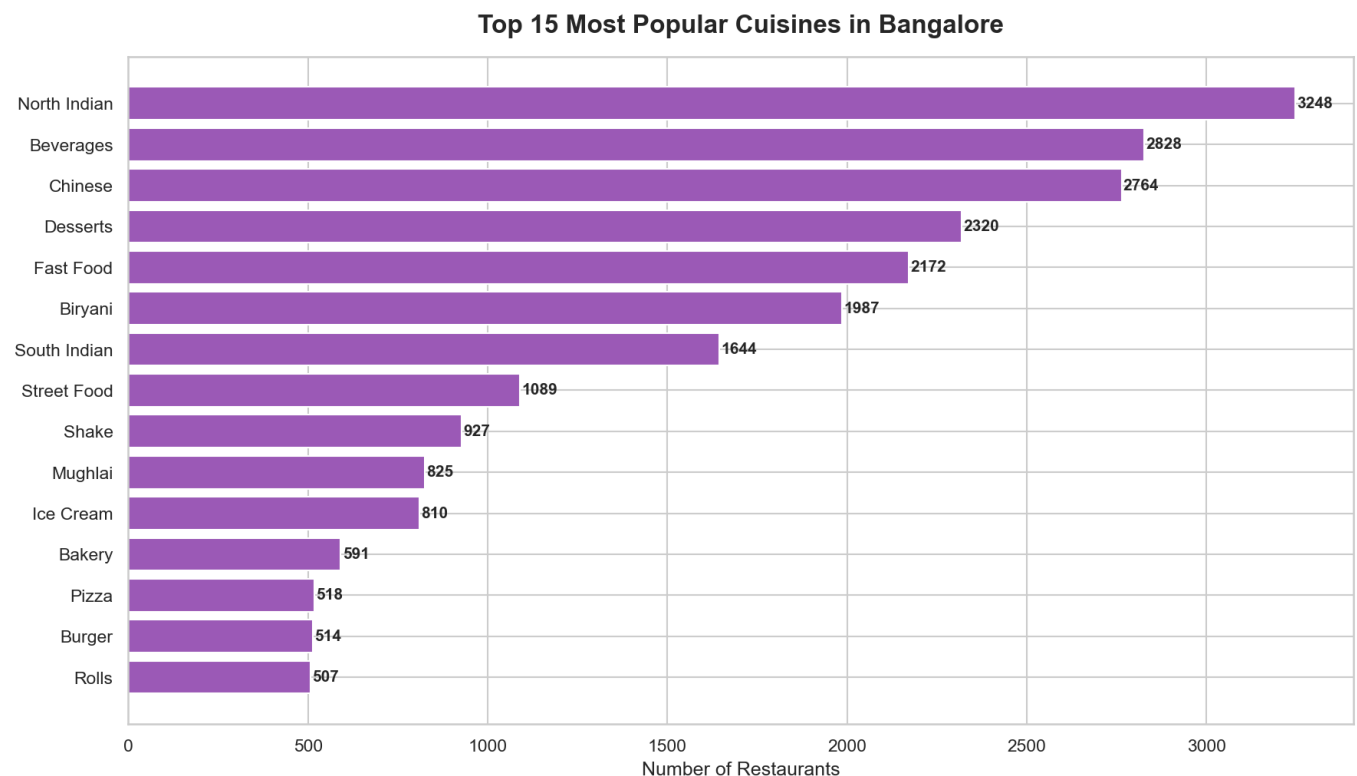
Restaurants by Rating Category



Cuisine Analysis

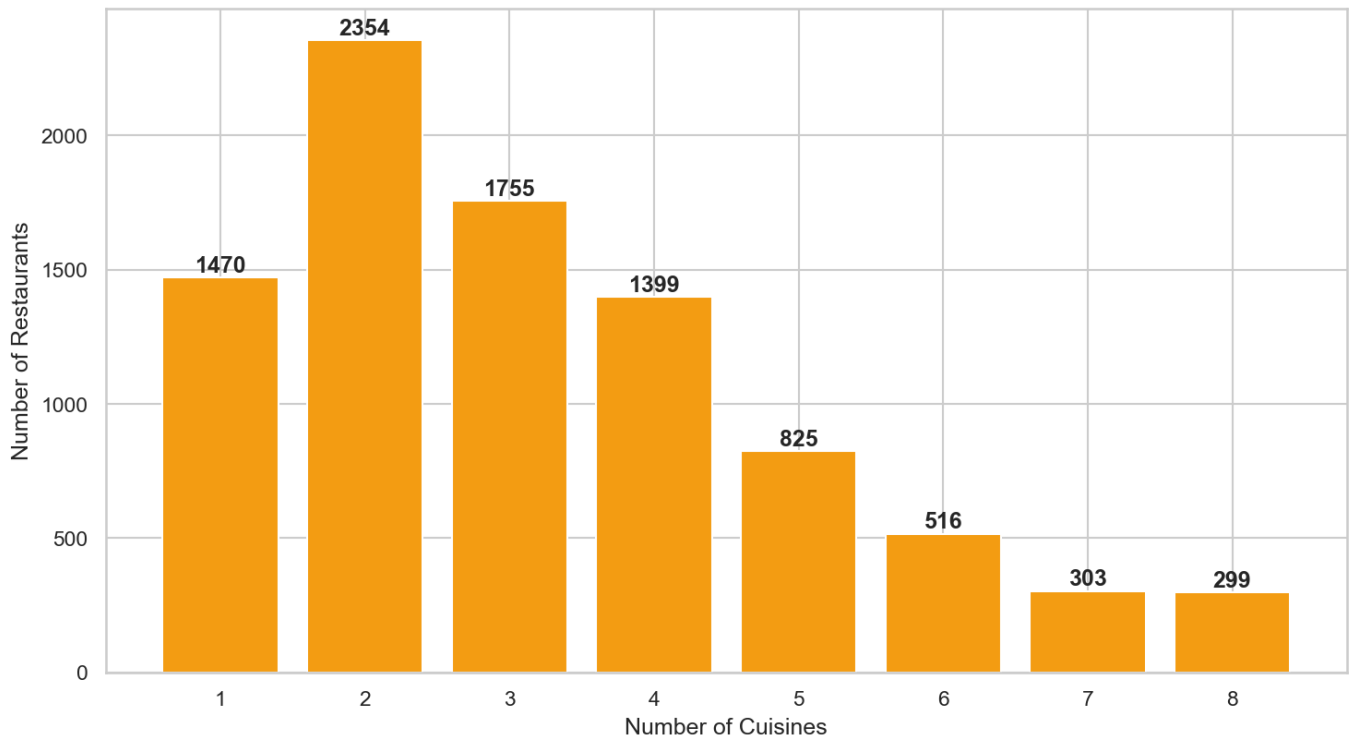
North Indian and Chinese cuisines dominate the Bangalore restaurant scene. Most restaurants serve multiple cuisines - the average restaurant offers 4-5 different cuisine types, showing high competition and diverse customer preferences.

Top 15 Most Popular Cuisines



Number of Cuisines per Restaurant

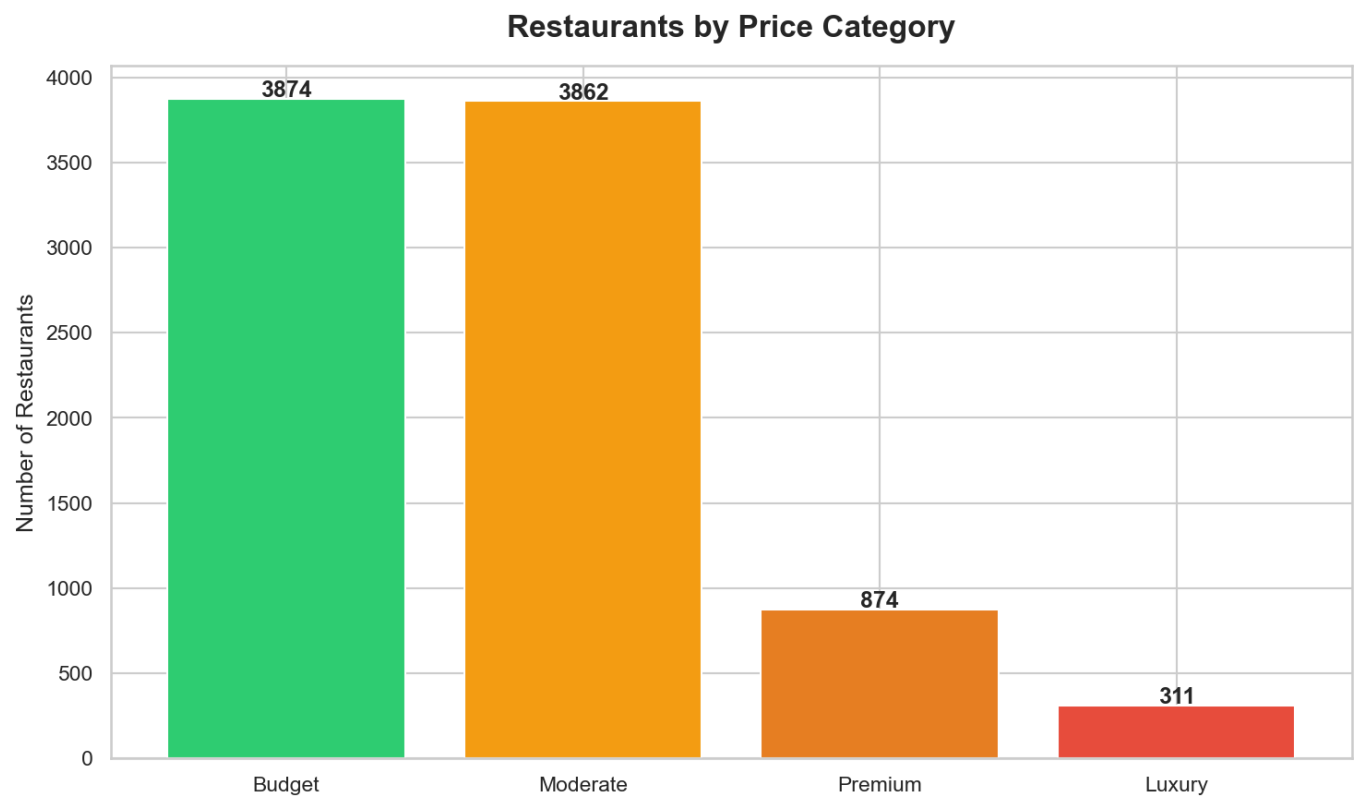
How Many Cuisines Does Each Restaurant Serve?



Price Analysis

The majority of Bangalore restaurants fall in the Budget (under Rs 200) and Moderate (Rs 200-500) price range. The average cost for two is Rs 340. Premium and Luxury restaurants tend to have slightly higher ratings.

Restaurants by Price Category



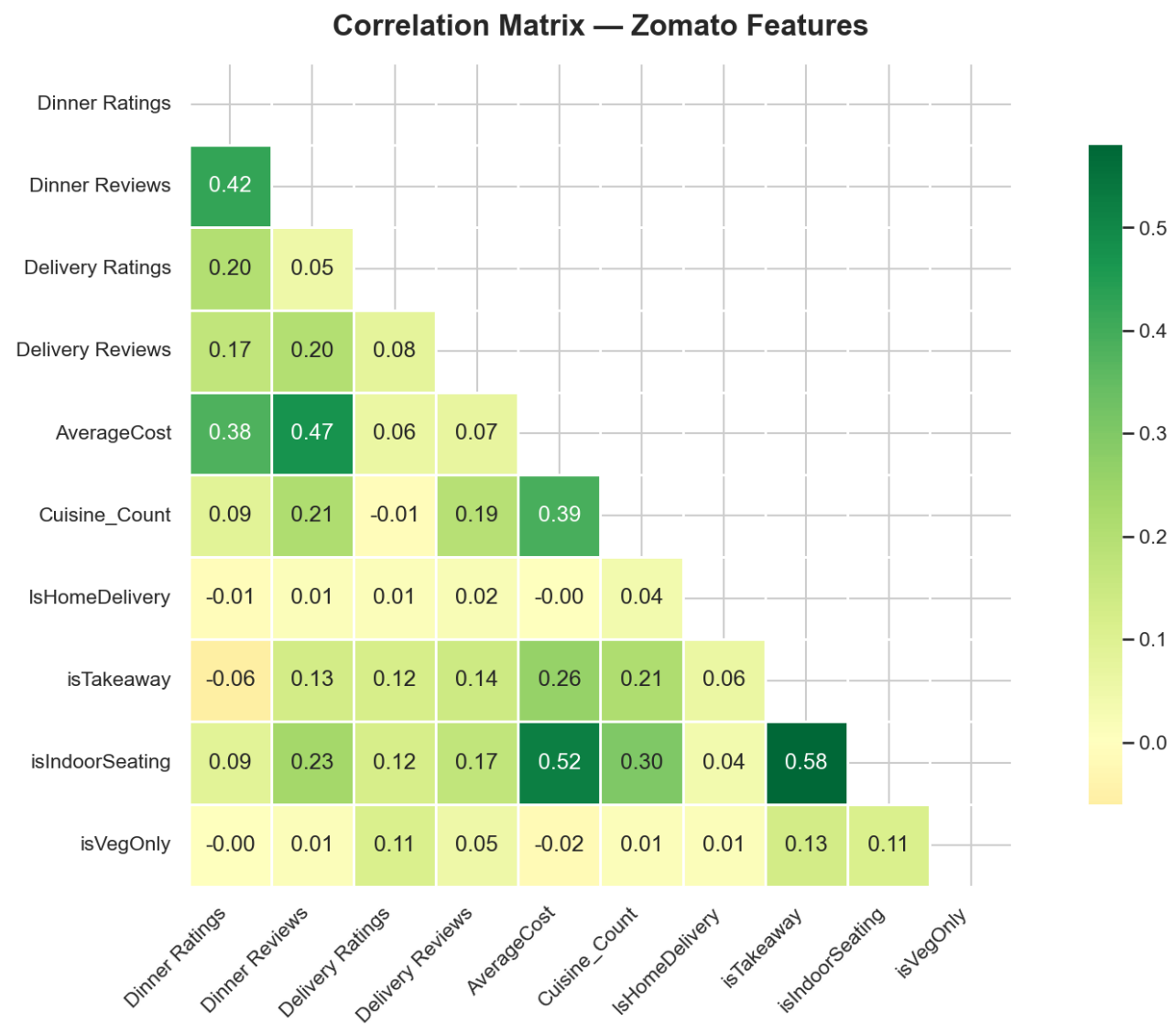
Rating Distribution by Price Category



Correlation Analysis

The correlation matrix reveals that Dinner Ratings and Delivery Ratings are strongly correlated - restaurants that perform well for dine-in also perform well for delivery. AverageCost shows a mild positive correlation with ratings.

Correlation Matrix - All Features



Key Business Insights

1. Location drives volume

Whitefield, BTM Layout and Koramangala have the most restaurants due to high IT workforce density. These are prime areas for new restaurant launches.

2. North Indian dominates

North Indian cuisine is served by the most restaurants, followed by Chinese and South Indian. Any new restaurant offering these cuisines faces high competition.

3. Delivery is mainstream

Over 60% of restaurants offer home delivery, showing that online food ordering is now a standard expectation in Bangalore.

4. Budget segment is largest

Most restaurants target the budget and moderate price segment (under Rs 500 for two), catering to the large working population.

5. Rating consistency

Dinner and delivery ratings are strongly correlated - a restaurant that delivers quality dine-in experience also delivers quality food, suggesting that food quality is the primary driver of ratings.

6. Veg-only is a niche

Only a small percentage of restaurants are veg-only, suggesting most restaurants cater to both veg and non-veg customers to maximize their customer base.